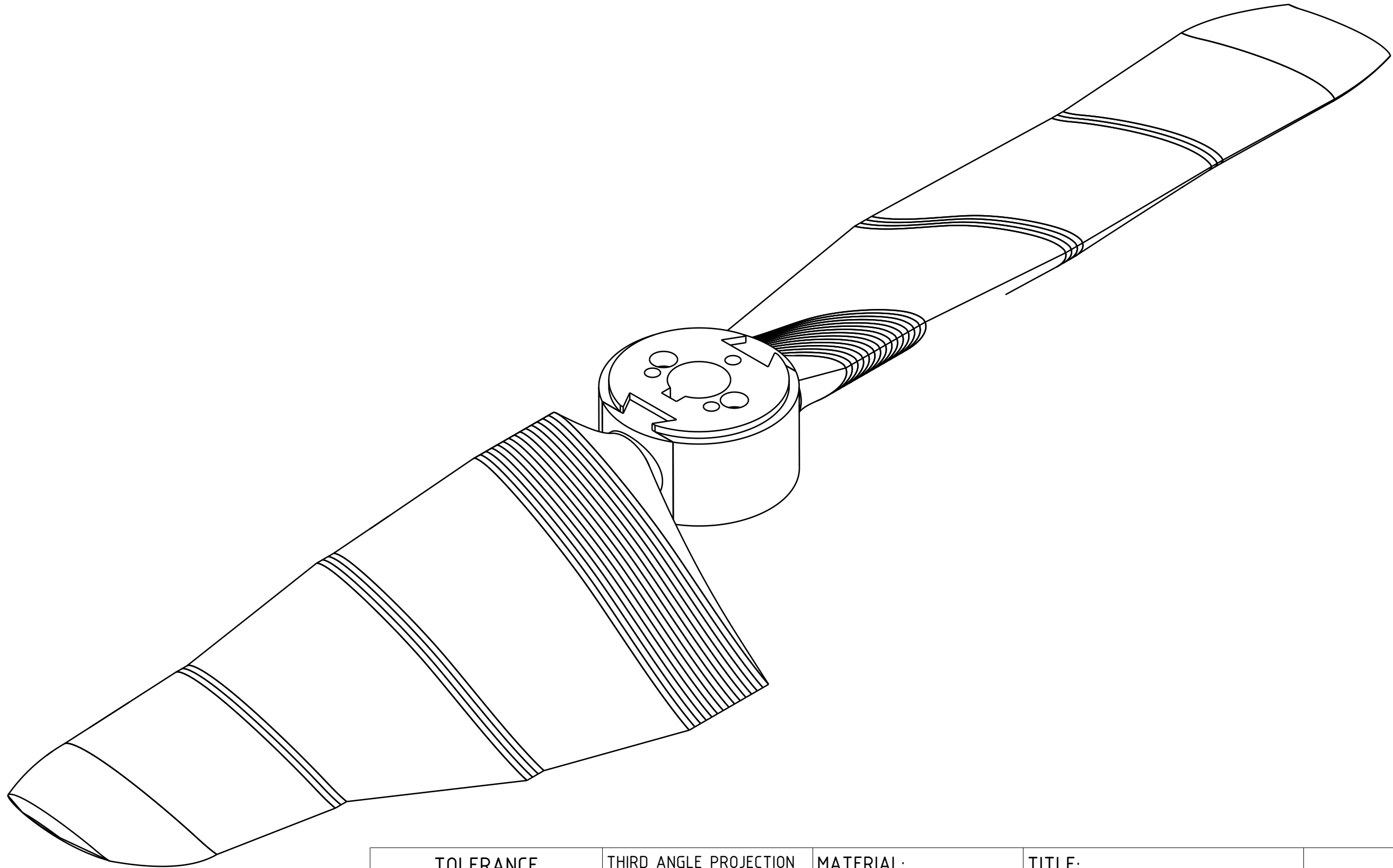
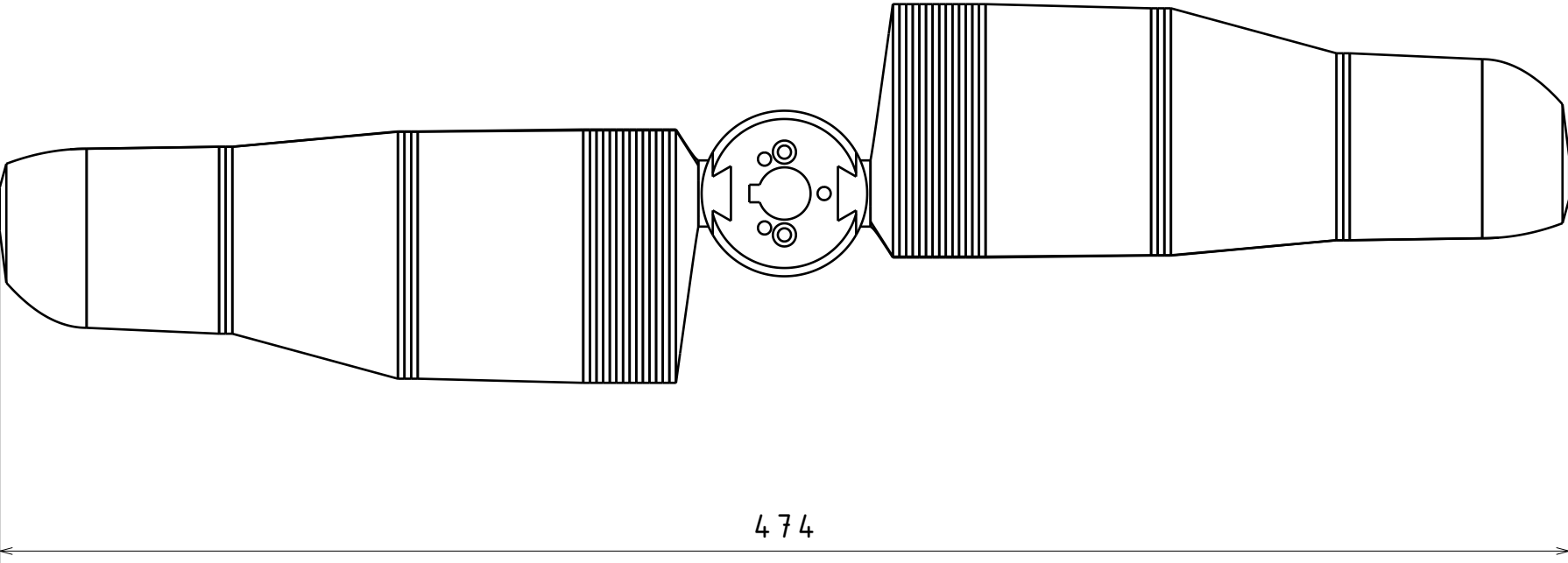




TOLERANCE		THIRD ANGLE PROJECTION		MATERIAL:		TITLE:		Imperial College London	
x=±0.5 x.x=±0.1 x.xx=±0.02	ANGULAR ±1°			DO NOT SCALE DRAWING		[*]			
	NAME	DATE		ALL DIMENSIONS ARE IN MILLIMETERS		DRAWING NUMBER:		DEPARTMENT OF AERONAUTICS	
DRAWN	[*]	[*]				[*]			
CHECKED	[*]	[*]							
APPROVED									
				A3		Scale [*]			



TOLERANCE		THIRD ANGLE PROJECTION		MATERIAL:		TITLE:		Imperial College London	
x=±0.5 x.x=±0.1 x.xx=±0.02	ANGULAR ±1°			DO NOT SCALE DRAWING		[*]			
	NAME	DATE		ALL DIMENSIONS ARE IN MILIMETERS		DRAWING NUMBER:		DEPARTMENT OF AERONAUTICS	
DRAWN	G19	29/05/2026				[*]			
CHECKED	G19	[*]							
APPROVED	G19								
				A3	Scale [*]			SHEET X OF X	REVISION
									[*]

PART	MATERIAL	PART DRAWING NUMBER	QUANTITY
W1	Plywood (2 ply)	AERO40004-DMT-G19-003	14
W2	Plywood (2 ply)	AERO40004-DMT-G19-005	3
W3	Plywood (2 ply)	AERO40004-DMT-G19-007	2
F1	Styrofoam	AERO40004-DMT-G19-004	1
F2	Styrofoam	AERO40004-DMT-G19-006	1
F3	Styrofoam	AERO40004-DMT-G19-008	1
P1	PLA	AERO40004-DMT-G19-002	1
P2	PLA	AERO40004-DMT-G19-009	1
CF1	Carbon Fibre Strip	AERO40004-DMT-G19-010	1
CF2	Carbon Fibre Strip	AERO40004-DMT-G19-011	1

ALPHABETS DENOTE MATERIAL:

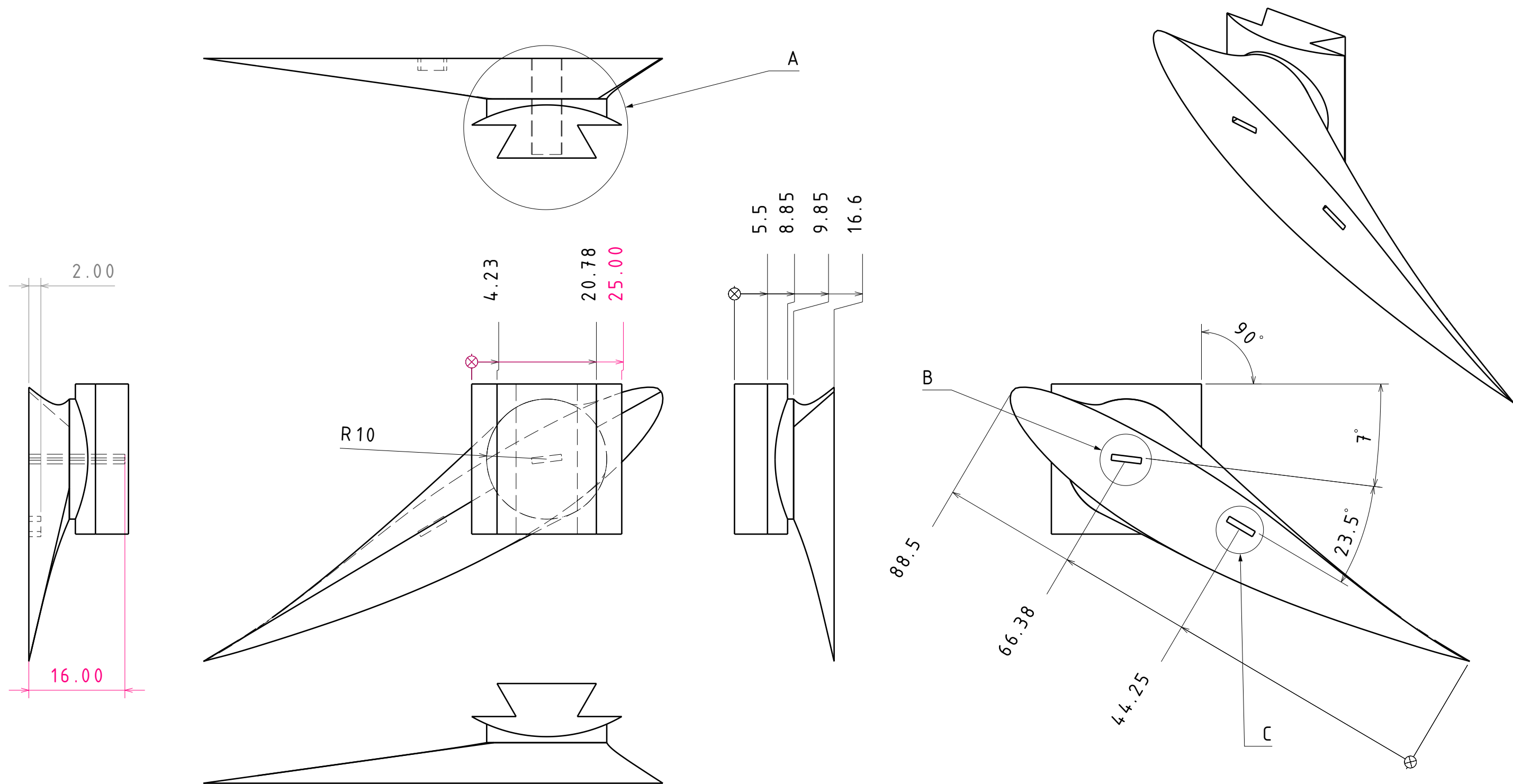
W: WOOD (PLYWOOD)
P: PLA
F: FOAM
CF: CARBON FIBRE

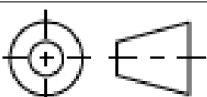
NUMBERS AFTER ALPHABET
DIFFERENTIATE PARTS OF SAME MATERIAL

TOLERANCE		THIRD ANGLE PROJECTION	MATERIAL:		TITLE:	
x=±0.5 x.x=±0.1 x.xx=±0.02	ANGULAR ±1°		DO NOT SCALE DRAWING		AE1 EP1 G19 DMT BLADE PROTOTYPE	
	NAME	DATE			DEPARTMENT OF AERONAUTICS	
DRAWN	G19	29/05/2026	ALL DIMENSIONS ARE IN MILIMETERS		DRAWING NUMBER:	
CHECKED	G19	29/05/2026			AERO40004-DMT-G19-001	
APPROVED	G19	29/05/2026				
			A3	Scale 1:1	SHEET 1 OF 2	
					REVISION A.1	

Imperial College London

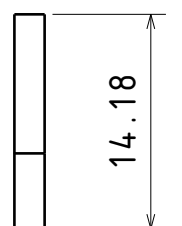
DEPARTMENT OF AERONAUTICS



TOLERANCE		THIRD ANGLE PROJECTION		MATERIAL:		TITLE:		Imperial College London	
x=±0.5 x.x=±0.1 x.xx=±0.02	ANGULAR ±1°			PLA		P1 - PLA ROOT			
	NAME	DATE		DO NOT SCALE DRAWING				DEPARTMENT OF AERONAUTICS	
DRAWN	G19	29/05/2026		ALL DIMENSIONS ARE IN MILLIMETERS		DRAWING NUMBER:			
CHECKED	G19	29/05/2026				AERO40004-DMT-G19-002			
APPROVED	G19	29/05/2026		A3 Scale 3:2				SHEET 1 OF 2	REVISION A.1

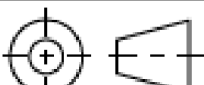
A diagram showing a rectangular plate (solid black) positioned on a circular base (dashed line). A dashed line extends from the center of the circle through the plate. An angle of 90° is indicated between a vertical dashed line and the line passing through the center. Another angle of 42.01° is indicated between a horizontal dashed line and the line passing through the center.

Rec1 and Rec2 are dimensionally identical
SG6040 Aerofoil Section



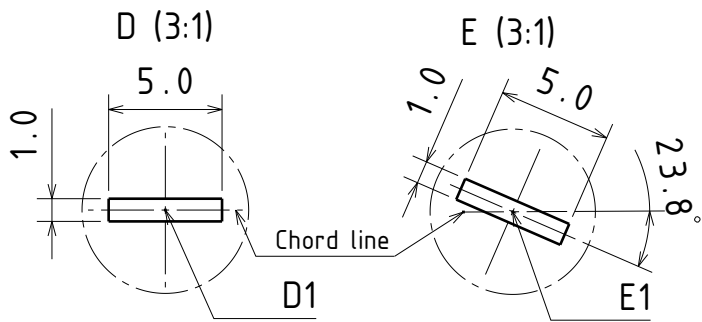
(2:1) 2

A diagram showing a 2:1 elliptical orbit. The orbit is represented by a dashed ellipse. Two objects, depicted as small rectangular blocks, are shown at different positions along the orbit. The text "2:1" is placed near the top of the orbit, indicating the orbital period ratio.

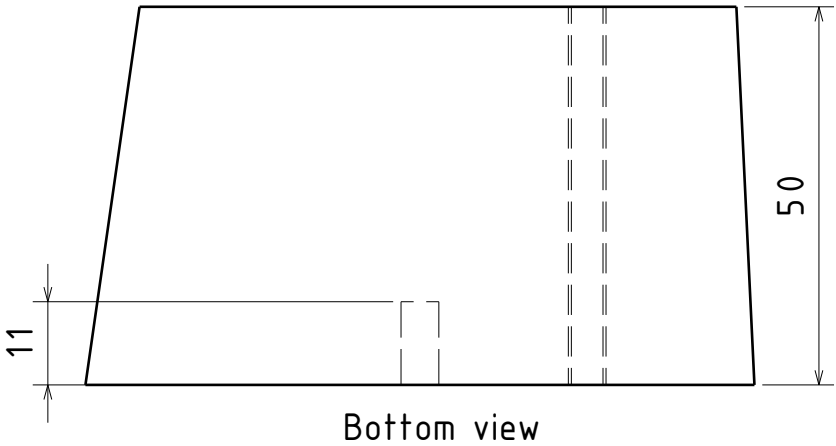
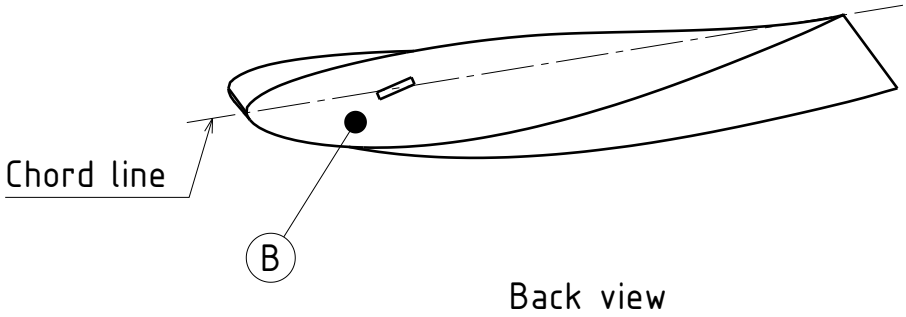
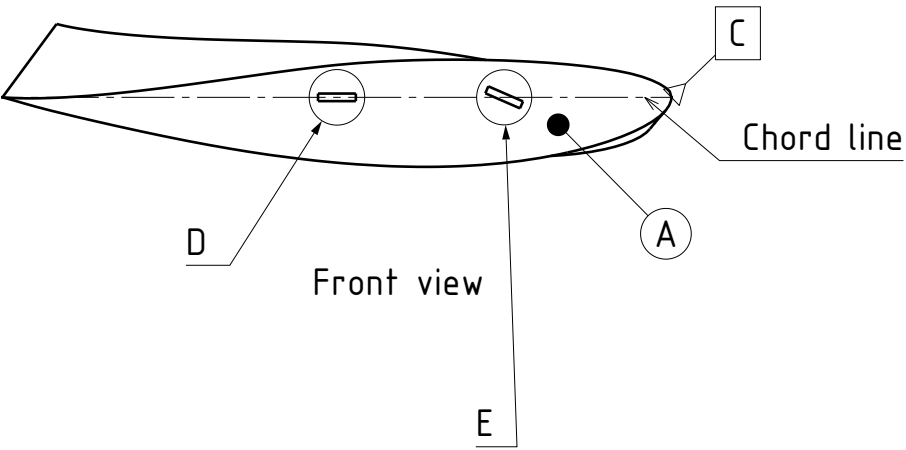
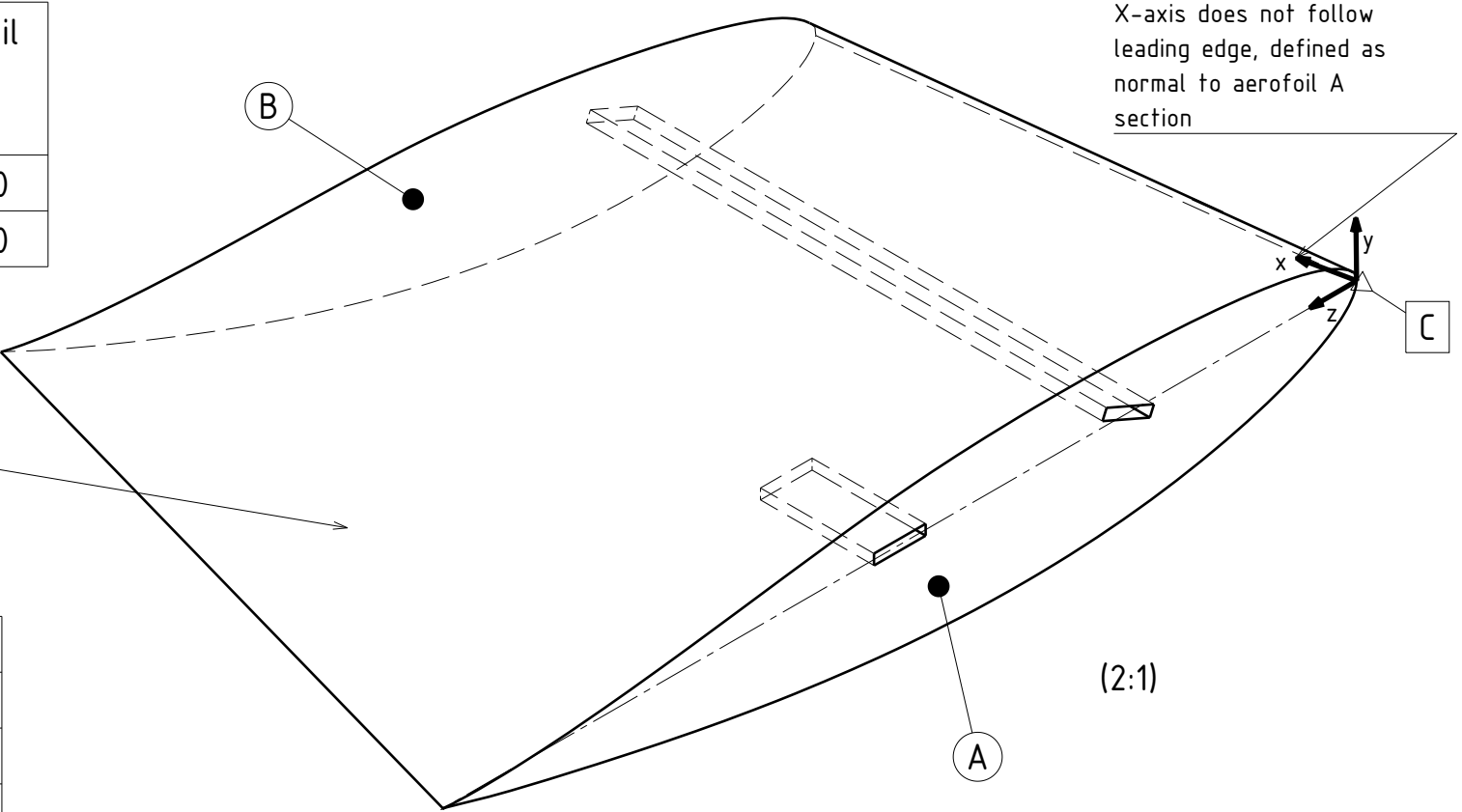
TOLERANCE		THIRD ANGLE PROJECTION		MATERIAL:		TITLE:		Imperial College London	
x=±0.5 x.x=±0.1 x.xx=±0.02	ANGULAR ±1°			PLYWOOD		W1 WOODEN AEROFOIL SECTION			
				DO NOT SCALE DRAWING					
	NAME	DATE		ALL DIMENSIONS ARE IN MILIMETERS		DRAWING NUMBER:		DEPARTMENT OF AERONAUTICS	
DRAWN	G19	29/05/2026				AERO40004-DMT-G19-003			
CHECKED	G19	29/05/2026							
APPROVED	G19	29/05/2026		A3 Scale: 2:1				SHEET 1 OF 1 REVISION 1	

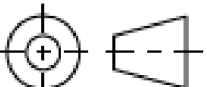
Aerofoil Section	Leading Edge Position (x,y,z from C) / mm	Local Chord / mm	Local Twist (around x-axis, positive anticlockwise) / °	Offset (chord fraction)	Aerofoil
A	0,0,0	88.5	0.0	0.25	SG6040
B	50,-3.997,0.417	80.0	-9.3	0.25	SG6040

Surface defined by linear ruled lofts between aerofoil sections



Position from C / mm ± 0.010			
Point	x	y	z
D1	0	0	44.250
E1	0	0	22.125

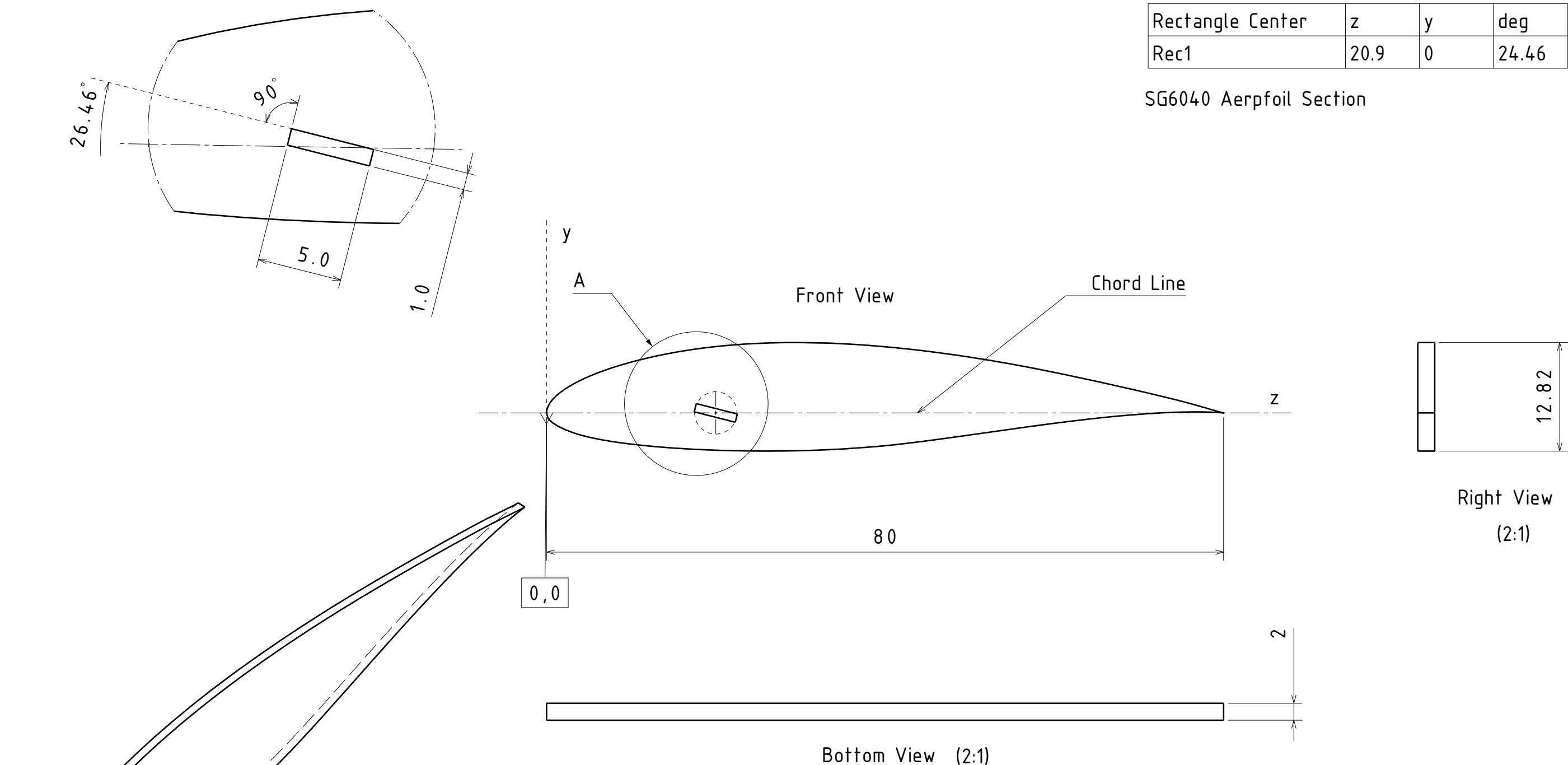


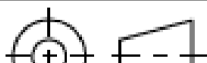
TOLERANCE		THIRD ANGLE PROJECTION		MATERIAL:		TITLE:		Imperial College London	
x=±0.5 x.x=±0.1 x.xx=±0.02	ANGULAR ±1°			STYROFOAM		F1 Styrofoam Section			
				DO NOT SCALE DRAWING					
	NAME	DATE		ALL DIMENSIONS ARE IN MILLIMETERS		DRAWING NUMBER: AERO40004-DMT-G19-004		DEPARTMENT OF AERONAUTICS	
DRAWN	G19	29/05/2026							
CHECKED	G19	29/05/2026		A3 Scale 1:1					
APPROVED	G19	29/05/2026							
								SHEET 1 OF 1	REVISION 1

A (4:1)

Rectangle Center	z	y	deg
Rec1	20.9	0	24.46

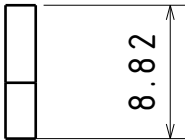
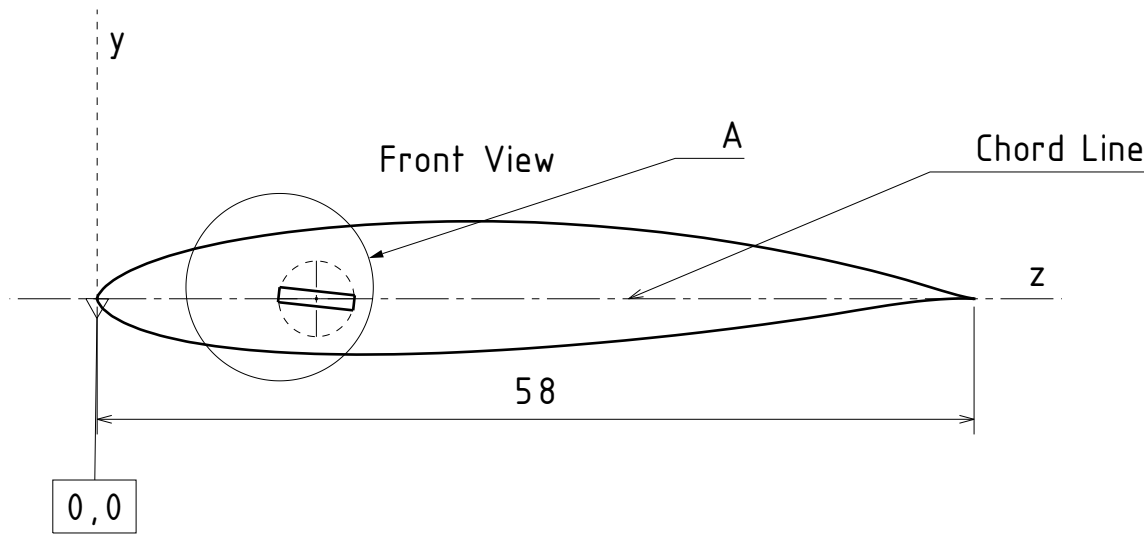
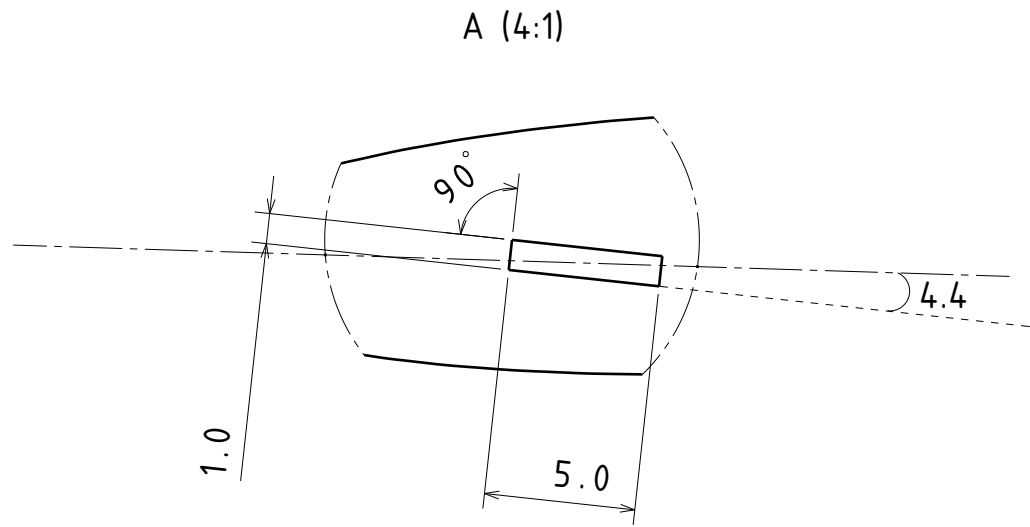
SG6040 Aerfoil Section



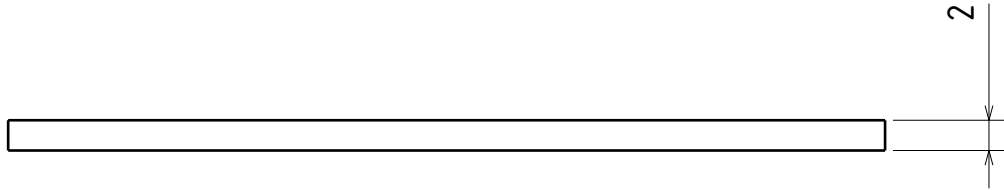
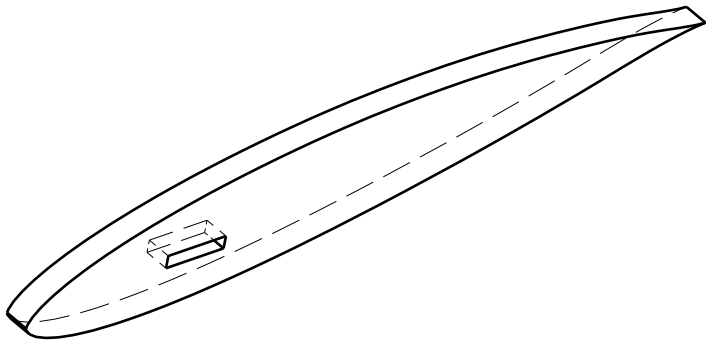
TOLERANCE		THIRD ANGLE PROJECTION		MATERIAL:		TITLE:		Imperial College London	
x=±0.5 x.x=±0.1 x.xx=±0.02	ANGULAR ±1°			PLYWOOD		W2 WOODEN AEROFOIL SECTION			
	NAME	DATE		DO NOT SCALE DRAWING		ALL DIMENSIONS ARE IN MILIMETERS		DEPARTMENT OF AERONAUTICS	
DRAWN	G19	29/05/2026							
CHECKED	G19	29/05/2026		A3		Scale: 2:1		DRAWING NUMBER: AERO40004-DMT-G19-005	
APPROVED	G19	29/05/2026							
								SHEET 1 OF 1	
								REVISION 1	

Rectangle Center	z	y	deg
Rec1	14.431	0	4.4

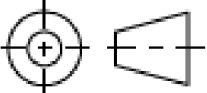
S838 Aerofoil Section



Right View (2:1)

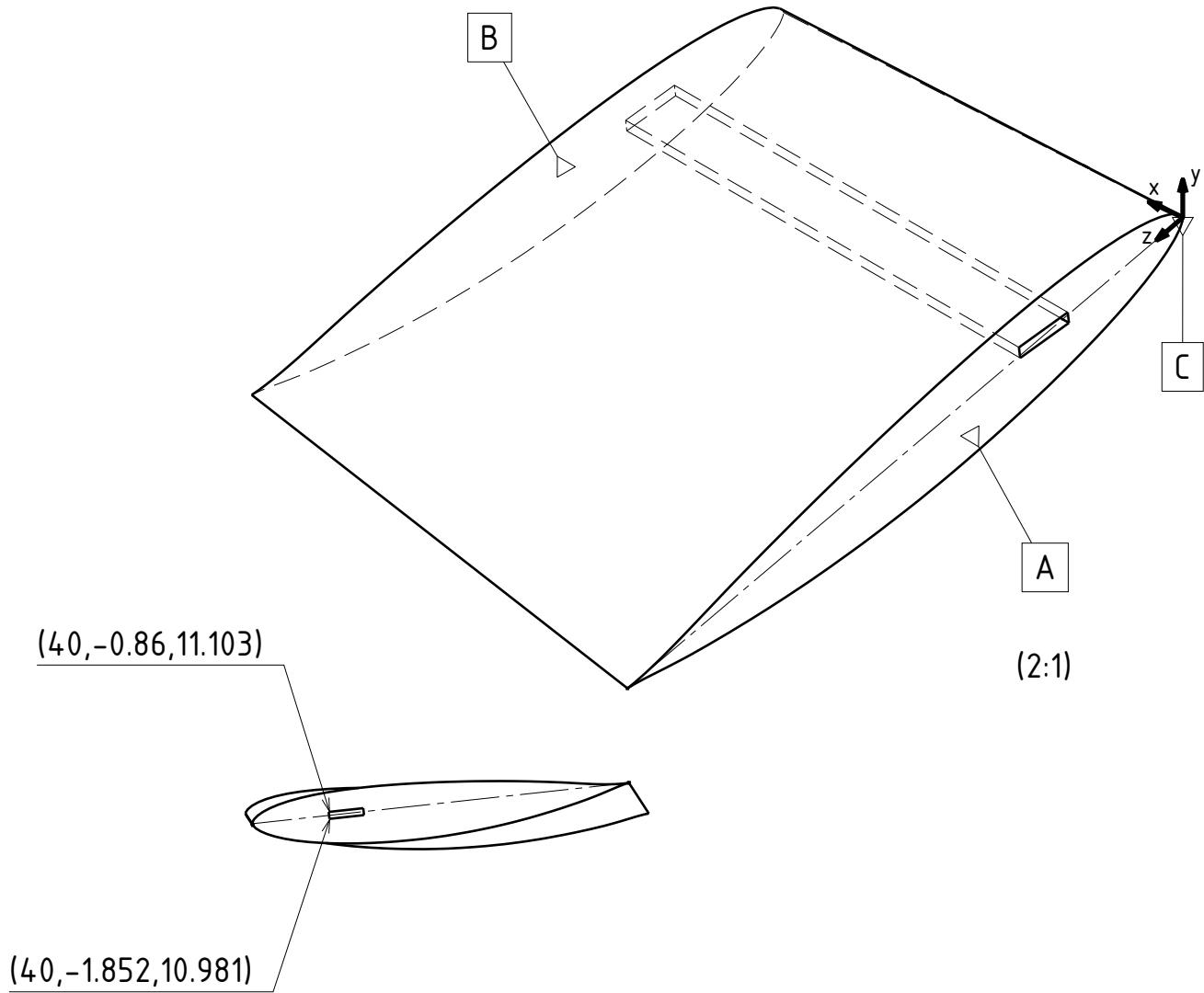
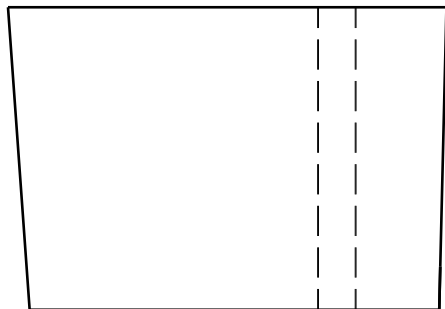
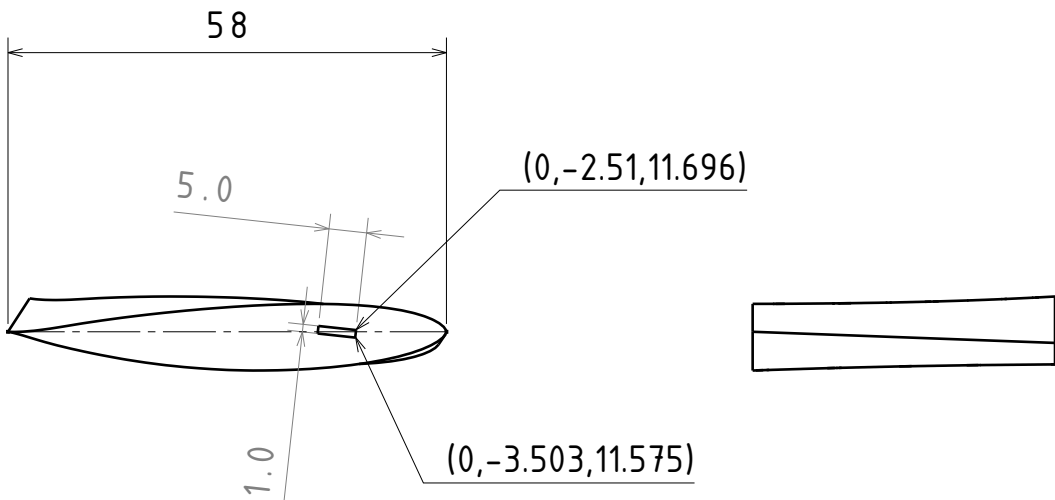


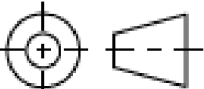
Bottom View (2:1)

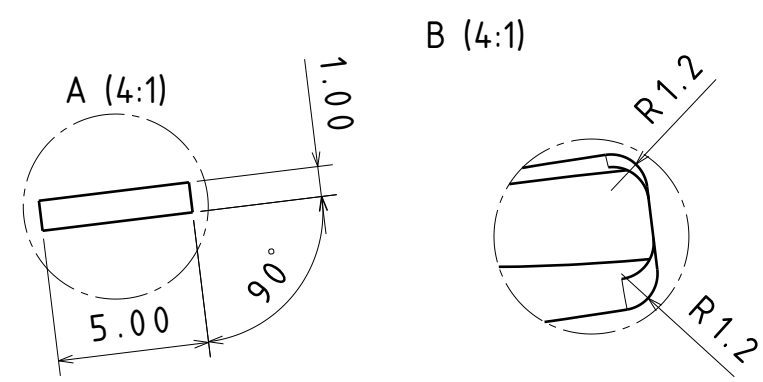
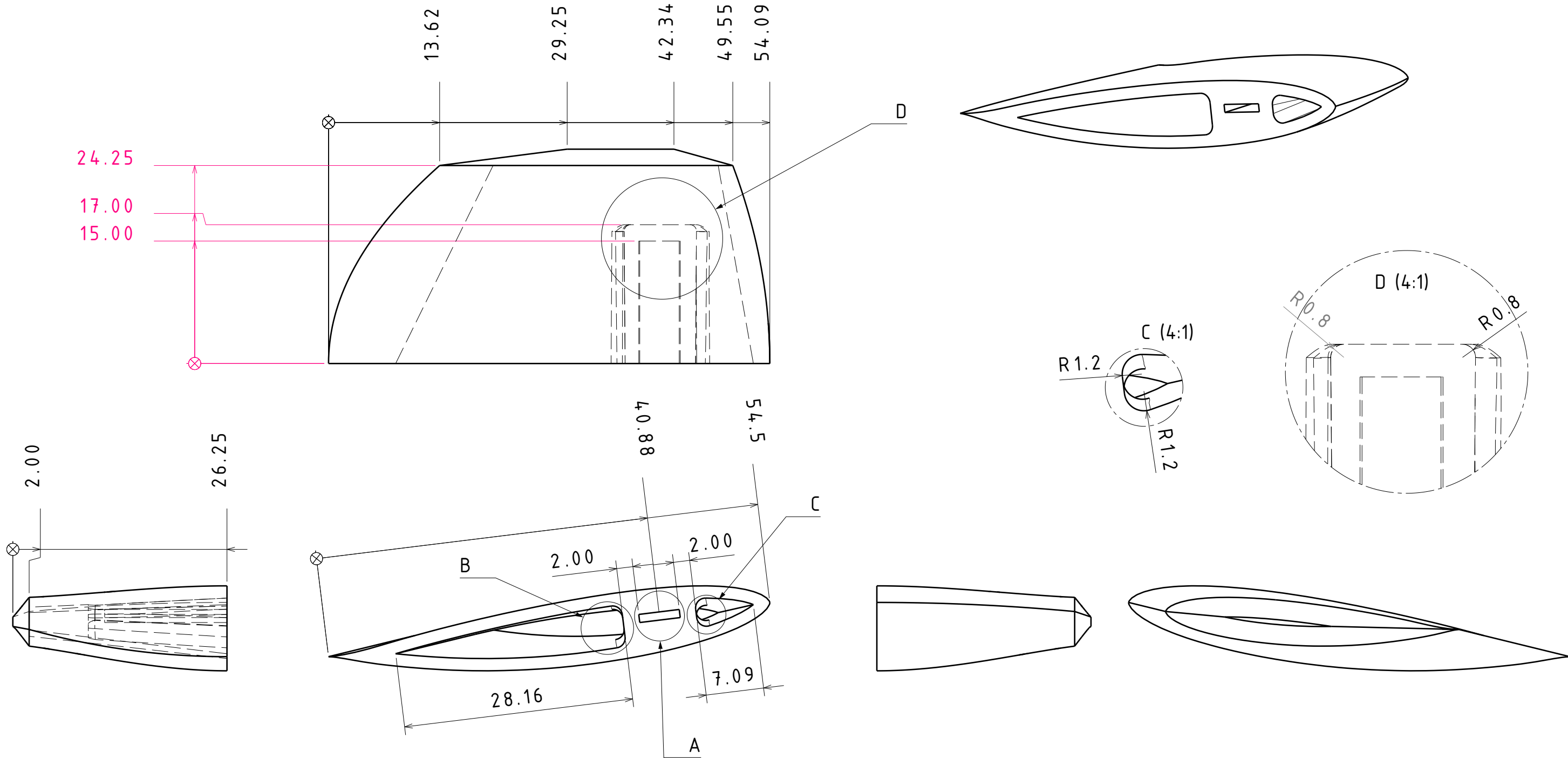
TOLERANCE		THIRD ANGLE PROJECTION		MATERIAL:		TITLE:		Imperial College London	
x=±0.5 x.x=±0.1 x.xx=±0.02	ANGULAR ±1°			PLYWOOD		W3 WOODEN AEROFOIL SECTION			
				DO NOT SCALE DRAWING					
	NAME	DATE		ALL DIMENSIONS ARE IN MILIMETERS		DRAWING NUMBER: AERO40004-DMT-G19-007		DEPARTMENT OF AERONAUTICS	
DRAWN	G19	29/05/2026							
CHECKED	G19	29/05/2026		A3 Scale: 2:1					
APPROVED	G19	29/05/2026						SHEET 1 OF 1	

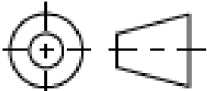
AEROFOIL SECTION	LE COORDINATE FROM C (x,y,z) /mm	LOCAL CHORD	OFFSET	LOCAL TWIST	AIRFOIL
A	(0,0,0)	58	0.25	0	S834
B	(40,-1.651,0.593)	54.5	0.25	6.2	S834

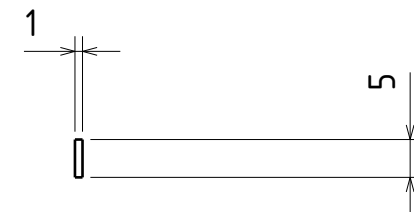
THIS SECTION IS DEFNIED BY LINEAR LOFTING OF THE TWO AIRFOIL

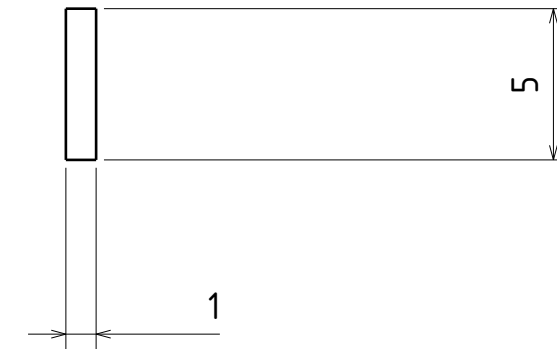
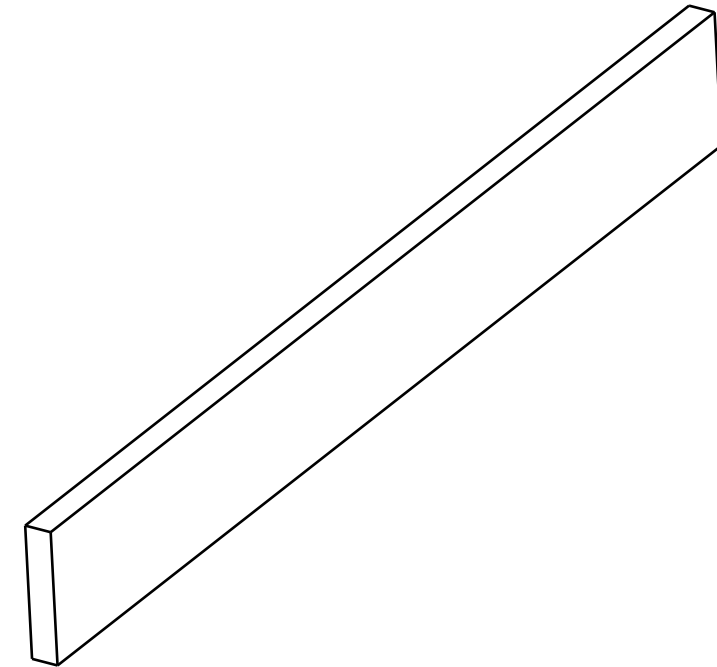


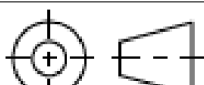
TOLERANCE		THIRD ANGLE PROJECTION		MATERIAL:		TITLE:		Imperial College London	
x=±0.5 x.x=±0.1 x.xx=±0.02	ANGULAR ±1°			STYROFOAM		F2 STYROFOAM SECTION 3			
				DO NOT SCALE DRAWING					
	NAME	DATE		ALL DIMENSIONS ARE IN MILLIMETERS		DRAWING NUMBER: AERO40004-DMT-G19-008		DEPARTMENT OF AERONAUTICS	
DRAWN	G19	29/05/2026							
CHECKED	G19	29/05/2026		A3	Scale 1:1				
APPROVED	G19	29/05/2026						SHEET 1 OF 1	



TOLERANCE		THIRD ANGLE PROJECTION		MATERIAL:		TITLE:		Imperial College London	
x=±0.5 x.x=±0.1 x.xx=±0.02	ANGULAR ±1°			PLA		P2 - PLA TIP			
				DO NOT SCALE DRAWING					
	NAME	DATE		ALL DIMENSIONS ARE IN MILLIMETERS		DRAWING NUMBER:		DEPARTMENT OF AERONAUTICS	
DRAWN	G19	29/05/2026				AERO40004-DMT-G19-009			
CHECKED	G19	29/05/2026							
APPROVED	G19	29/05/2026		A3 Scale 2:1				SHEET 1 OF 1	
								REVISION A.1	





TOLERANCE		THIRD ANGLE PROJECTION		MATERIAL:		TITLE:		Imperial College London	
x=±0.5 x.x=±0.1 x.xx=±0.02	ANGULAR ±1°			CARBON FIBRE		CARBON FIBRE ROD 2			
				DO NOT SCALE DRAWING					
	NAME	DATE	ALL DIMENSIONS ARE IN MILIMETERS		DRAWING NUMBER:		DEPARTMENT OF AERONAUTICS		
DRAWN	G19	29/05/2026			AERO40004-DMT-G19-011				
CHECKED	G19	29/05/2026							
APPROVED	G19	29/05/2026	A3 Scale 2:1				SHEET 1 OF 1 REVISION A.1		